

SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 05 : Introduction to Copulas and Applications

W5L1 : Preliminary Concepts of Copula and Sklar's Theorem

BY

Prof. Rajib Maity
Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

Preliminary Concepts of Copula and Sklar's Theorem

NPTEL

Introduction

- The word Copula originates from a Latin word 'copulare', which means 'to join together'.
- In many cases of statistical modeling, Copula is essential to obtain the joint probability distribution function between two or more random variables.
- Even though the marginal distributions of each of the random variables are known, their joint distributions may not be easy to derive from these marginal distributions.
- However, copula can be used to obtain their joint distribution, if the information on scale-free measures of dependence between random variables is available.

Preliminary Concepts

Definition of Copula

- Let X and Y be a pair of random variables with Cumulative Distribution Functions (CDF) $F_X(x)$ and $F_Y(y)$ respectively and their joint CDF be $H_{X,Y}(x,y)$.
- Hence, each pair (x,y) of real numbers mapped to a point $F_X(x)$, $F_Y(y)$ in the unit square, i.e., I^2 or [0,1] × [0,1].
- The ordered pair is associated with a value $H_{X,Y}(x,y)$ that lies within the interval [0,1], which is defined as Copula.
- Thus, copula (designated by C), is a function that joins or couples one-dimensional marginal distributions of multiple random variables to their joint distribution function.

Preliminary Concepts

- Any form of marginal distributions can be coupled to get their joint distribution which is the reason for the popularity of copula theory in many areas of research.
- Moreover, the theory of copula can be extended to higher dimensions also.

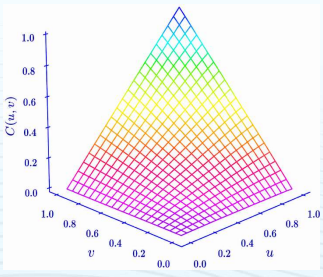
Graphical Representation of Copula

- An n-dimensional copula is represented in a I^n-dimensional space.
- Graphically, only two-dimensional copulas can be shown as a surface in I^3 space or as a contour plot in I^2.

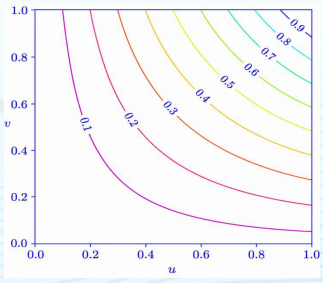
Preliminary Concepts

Graphical Representation of Copula

For example, a two-dimensional independent copula, given by $C(u, v) = uv$ for $0 \leq u, v \leq 1$, is graphically represented in the following figure:



3-D Surface plot



Contour plot

Sklar's Theorem

- Application of copula to probability and statistics is achieved through Sklar's theorem.
- It states that, if $H_{X,Y}(x,y)$ is a joint distribution function, then there exists a copula $C(u,v)$ such that, for all $x,y \in \mathbb{R}$,

$$H_{X,Y}(x,y) = C(F_X(x), F_Y(y))$$

where $F_X(x)$ and $F_Y(y)$ are the marginal distribution of X and Y respectively.

- Hence, if $F_X(x)$ and $F_Y(y)$ are continuous, then copula function is unique, otherwise copula is uniquely determined on $Ran(F_X(x)) \times Ran(F_Y(y))$, where $\mathrm{Ran}(\bullet)$ represents range for $\bullet$.

Sklar's Theorem

- Sklar's theorem is used for coupling two marginal distributions to obtain their joint distribution.
- The equation (1) can be inverted as:

$$C(u,v) = H\left(F_X^{(-1)}(u), F_Y^{(-1)}(v)\right)$$

where $F_X^{(-1)}(u)$ and $F_Y^{(-1)}(v)$ are called '*quasi-inverse*' of $F_X(x)$ and $F_Y(y)$ respectively.

- If any marginal distribution ($F_X(x)$ and $F_Y(y)$) is strictly increasing function, then the quasi-inverse is same as its inverse (denoted as $F_X^{(-1)}$ and $F_Y^{(-1)}$).

Sklar's Theorem

- However, if any marginal distribution $F_X(x)$ is not strictly increasing, then the quasi-inverse is given by:
  - For any $t \in Ran(F_X(x)), F_X^{(-1)}(t)$ is any number $x \in \mathbb{R}$, such that $F_X(x) = t$, i.e., for all t in $Ran(F_X(x))$,

$$F_X\left(F_X^{(-1)}(t)\right) = t$$

  - If t is not in $Ran(F_X(x))$, then

$$F_X^{(-1)}(t) = \inf\{x | F_X(x) \geq t\} = \sup\{x | F_X(x) \leq t\}$$

where inf stands for infimum and sup stands for supremum.

Sklar's Theorem

- If $F_X(x)$ and $F_Y(y)$ are continuous functions, then $C(u, v)$ is a valid copula function.
- Hence, copula can be constructed using the information of marginal distributions and joint distribution.
- This method of constructing copula is called inversion technique to construct copula.

Example

The joint distribution function describing the dependence between two random variables X and Y is given by:

$$H_{X,Y}(x,y) = \begin{cases} \frac{(x+1)(e^y-1)}{x+2e^y-1} & (x,y) \in [-1,1] \times [0,\infty) \\ 1-e^{-y} & (x,y) \in [1,\infty) \times [0,\infty) \\ 0 & elsewhere \end{cases}$$

The marginal distribution functions of X and Y are

$$F_X(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{2} & x \in [-1,1] \\ 0 & x > 1 \end{cases} \quad \text{and} \quad F_Y(y) = \begin{cases} 0 & y < 0, \\ 1-e^{-y} & y \geq 0 \end{cases}$$

Find the corresponding copula function.

Solution

If the reduced variate of X is $u \in [0,1]$ (i.e., $F_X(x) = u$ or $F_X^{(-1)}(u) = x$), then the quasi-inverse of $F_X(x)$ is given by

$$u = \frac{F_X^{(-1)}(u) + 1}{2}$$

or $$F_X^{(-1)}(u) = 2u - 1$$

Similarly, if v is the reduced variate of Y, then the quasi-inverse of $F_Y(y)$ is given by ($F_Y^{(-1)}(v) = y$)

$$v = 1 - e^{-F_Y^{(-1)}(v)}$$

$$e^{-F_Y^{(-1)}(v)} = 1 - v$$

$$F_Y^{(-1)}(v) = -\ln(1-v)$$

Solution

Both $F_X(x)$ and $F_Y(y)$ are continuous functions with range I. Hence, the corresponding copula C is given by

$$C = H_{X,Y}\left(F_X^{(-1)}(u), F_Y^{(-1)}(v)\right)$$

or, $$C = \frac{(2u - 1 + 1)(e^{-\ln(1-v)} - 1)}{2u - 1 + 2e^{-\ln(1-v)} - 1}$$

or, $$C = \frac{2u(\frac{1}{1-v} - 1)}{2u + \frac{2}{1-v} - 2}$$

Solution

or $$C = \frac{2u(\frac{1 - 1 + v}{1-v})}{\frac{2u - 2uv + 2 - 2 + 2v}{1-v}}$$

$$C = \frac{uv}{u + v - uv}$$

Thus, the corresponding copula function is , $C = \frac{uv}{u+v-uv}$


FREE ONLINE EDUCATION
swayam
Thank You!
alt
15



SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 05 : Introduction to Copulas and Applications

W5L2: Basic Properties of Copula Functions

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

Basic Properties of a Copula Functions

NPTEL

Mathematical Definition of Copula

- A two-dimensional copula is a function C having domain of I^2 and has following properties:

a) $\mathrm{Dom}\ C = S_1 \times S_2$, where S_1 and S_2 are subsets of I containing 0 and 1, i.e., $\mathrm{I} = [0,1]$.

b) C is grounded and 2-increasing.

c) For every u in S_1 and v in S_2,

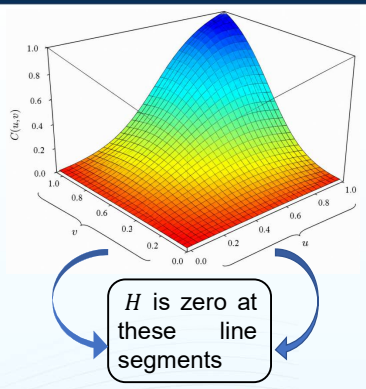
$$C(u,1) = u \text{ and } C(1,v) = v$$

NPTEL

Basic Terminologies

Grounded Function

- A two-dimensional function $G(x,y)$ in the domain $S_1 \times S_2$ is called grounded, if $G(a_1,y) = 0 = G(x,a_2)$ for all $(x,y) \in Dom(G(x,y))$, where, a_1 and a_2 are the lowest values of S_1 and S_2 respectively.



Pictorial representation of a two-dimensional grounded function

- Copula functions are required to be grounded. As domain of copula function is I^2, so $a_1 = a_2 = 0$.

- The copula function shown in figures are grounded as the value of copula function at u and v axes are zero, i.e., $C(0,v) = C(u,0) = 0$.

NPTEL

Basic Terminologies

2-increasing Function

- The concept of 2-increasing function in two-dimensional case is analogous to non-decreasing functions in one dimension.

H−volume

- Assume that S_1 and S_2 are two non empty sub-sets of set of real numbers, $\mathbb{R}$, and $H(x, y)$ is a function defined on $S_1 \times S_2$.
- For any two points (x_1, y_1) and (x_2, y_2) in $S_1 \times S_2$, the corresponding rectangle B can be defined as $[x_1, x_2] \times [y_1, y_2]$.
- The H −volume for rectangle B is the volume enclosed by H function and XY plane.
- Mathematically, *H*-volume for rectangle B is expressed as

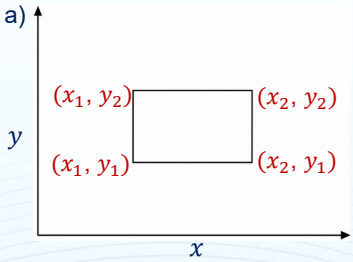
$$V_H(B) = H(x_2, y_2) - H(x_1, y_2) - H(x_2, y_1) + H(x_1, y_1)$$

Basic Terminologies:

2-increasing Function

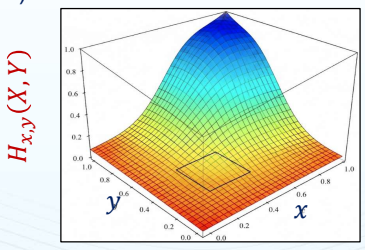
The concept of a 2-increasing function is graphically represented in the figure.

a)



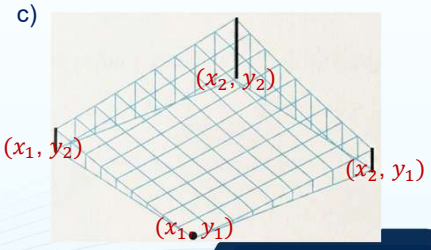
Rectangle B Denoted as $[x_1, y_1] \times [x_2, y_2]$

b)



Base area of the surface is the domain of H, Dom H

c)



H-volume of B denoted as $V_H(B)$

$$V_H(B) = H(x_2, y_2) - H(x_1, y_2) - H(x_2, y_1) + H(x_1, y_1)$$

Basic Terminologies

2-increasing Function

For every u_1, u_2, v_1, v_2 in I, if $u_1 \leq u_2$ and $v_1 \leq v_2$, then

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) \geq 0$$

This property indicates that the copula functions are 2-increasing.

- A two-dimensional real function is 2-increasing, if *H*-volume of any rectangle B is non-negative, i.e., $V_H(B) \geq 0$ for all rectangles B whose vertices lie in $Dom(H)$.

Frechet-Hoeffding Bounds

- Assume, C be a copula, then for every (u, v) in $Dom(C)$

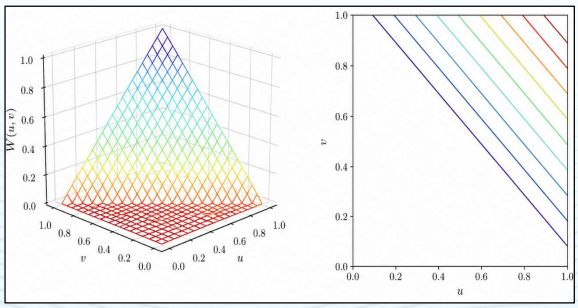
$$max(u + v - 1,0) \leq C(u, v) \leq \min(u, v)$$
$$W(u, v) \leq C(u, v) \leq M(u, v)$$

where $W(u, v) = max(u + v - 1,0)$ and $M(u, v) = \min(u, v)$.

- The functions $W(u, v)$ and $M(u, v)$ are called lower and upper Frechet-Hoeffding bounds respectively.
- As copulas couple two marginal distributions to obtain the joint distribution, Frechet-Hoeffding Bounds also apply as bounds to joint probability distributions.

Frechet-Hoeffding Bounds

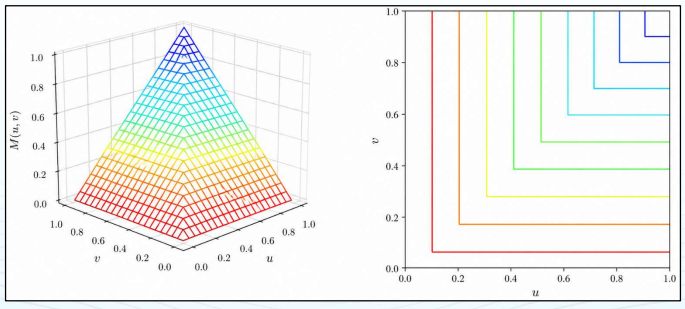
- Interestingly, for any $t \in \mathrm{I}$, these bounds define the region in I^2,for which copula/joint distribution function can have a value equal to t.
- The graphical representation for these bounds are given in the following figure.



a b

Graphical representation of $W(u,v)$ —a) 3-d representation b) Contour Plot

Frechet-Hoeffding Bounds



a b

Graphical representation of $M(u,v)$ —a) 3-d representation b) Contour Plot

Example

Determine whether each of the following candidate functions can satisfy as a valid copula function, where $u, v \in \mathrm{I}$.

a) $C(u,v) = uv$

b) $C(u,v) = 1$

c) $C(u,v) = \max(u+v-1, 0)$

d) $C(u,v) = \frac{u^2+v^2}{2}$

e) $C(u,v) = |u+v-2|$

Solution

a) For the function $C(u,v) = uv$, for all $u_1, u_2, v_1, v_2 \in \mathrm{I}$, such that $u_1 \leq u_2$ and $v_1 \leq v_2$,

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1)$$
$$= u_2\, v_2 - u_2\, v_1 - u_1\, v_2 + u_1\, v_1$$

Assuming $u_2 = u_1 + d$,

Then $u_2 v_2 + u_1 v_1 = (u_1 + d)v_2 + (u_2 - \mathrm{d})v_1$
$$= u_1 v_2 + u_2 v_1 + d(v_2 - v_1).$$

Hence, the above expression reduces to $\mathrm{d}(v_2 - v_1)$. As $d \geq 0$ and $(v_2 - v_1) \geq 0$, so

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) \geq 0$$

Solution

Then, for copula $C(u,v) = uv$,

$$C(u,0) = u \times 0 = 0$$
$$C(0,v) = 0 \times v = 0$$

and

$$C(u,1) = u \quad \text{and} \quad C(1,v) = v$$

Hence, the function given by $C(u,v) = uv$ is both grounded, 2-increasing in I . $C(u,v) = uv$ is a valid copula function.

Solution

b) For the function $C(u,v) = 1$, for any $u_1, u_2, v_1, v_2 \in \mathrm{I}$,

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) = 1 - 1 - 1 + 1 = 0$$

However, the function is not grounded on u and v axis as

$$C(u,0) = C(0,v) = 1 \neq 0$$

Moreover,

$$C(u,1) = 1 \neq u \quad \text{and,} \quad C(1,v) = 1 \neq v$$

Hence, the function given by $C(u,v) = 1$ is not a valid copula function

Solution

c) The function given by $C(u,v) = [\max(u+v-1,0)]$ should be 2-increasing in I. For all $u_1, u_2, v_1, v_2 \in$ I, such that $u_1 \leq u_2$ and $v_1 \leq v_2$, the function (by definition) is lower bounded by 0 and upper bounded by 1.

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) \geq 0$$

$$C(u_2, v_2) + C(u_1, v_1) \geq C(u_2, v_1) + C(u_1, v_2)$$

By definition of $C(u,v)$ no negative value of $C(u,v)$ is possible, so there are following four possibilities

i) $u_2 + v_2 \leq 1$ then $C(u_2, v_2) = C(u_1, v_1) = C(u_2, v_1) = C(u_1, v_2) = 0$ and

Hence $C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) = 0$

Solution

ii) $u_2 + v_1 \geq 1$ and $u_1 + v_2 \leq 1$ then $C(u_2, v_2) = u_2 + v_2 - 1 \geq 0$, $C(u_2, v_1) = u_2 + v_1 - 1 \geq 0$ and $C(u_1, v_1) = C(u_1, v_2) = 0$.

Hence, $C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1) = u_2 + v_2 - 1 - u_2 - v_1 + 1 = v_2 - v_1 \geq 0$

iii) $u_1 + v_2 \geq 1$ and $u_2 + v_1 \leq 1$ then $C(u_2, v_2) = u_2 + v_2 - 1 \geq 0$, $C(u_1, v_2) = u_1 + v_2 - 1 \geq 0$ and $C(u_1, v_1) = C(u_2, v_1) = 0$.

Hence, $C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1)$
$= u_2 + v_2 - 1 - u_1 - v_2 + 1 = u_2 - u_1 \geq 0$

Solution

iv) $u_1 + v_1 \leq 1$ then

$$C(u_2, v_2) - C(u_2, v_1) - C(u_1, v_2) + C(u_1, v_1)$$
$$= u_2 + v_2 - 1 - u_2 - v_1 + 1 - u_1 - v_2 + 1 + u_1 + v_1 - 1 = 0$$

Hence, the function given by $C(u, v) = \max(u + v - 1, 0)$ is 2-increasing.

The function is also grounded as $C(u, 0) = C(0, v) = 0$.

Furthermore, $C(u, 1) = \max(u + 1 - 1, 0) = u$ and similarly, $C(u, v) = \max(1 + v - 1, 0) = v$.

Hence, the $C(u, v)$ is a valid copula function.

Solution

d) The function given by $C(u, v) = \frac{u^2+v^2}{2}$ is not grounded as $C(u, 0) = \frac{u^2}{2} \neq 0$ for all $u \in \mathrm{I}$;

It should be noted that the violation of even a single copula property is sufficient to conclude that the function is not a valid copula.

e) The function $C(u, v) = |u + v - 2|$ is not grounded as $C(u, 0) = |u - 2|$,

Hence it can not be a valid copula function.


FREE ONLINE EDUCATION
swayam
Thank You!



SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 05 : Introduction to Copulas and Applications

W5L3 : Non-parametric Measures of Association

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

- **Non-parametric, Scale-free Measures of Association**

NPTEL

Non-parametric Measures of Association

- In many practical applications, it is important to assess the degree of association between two variables.
- The relationship between variables may be linear or non-linear, and the form of the relationship may not always be known in advance.
- Therefore, measures of associations that can suggest the existence of association between two variables irrespective of the nature of their inter-relationship are needed.

Non-parametric Measures of Association

- One way to achieve this is by measuring the association between the ranks of the variables, instead of their values.
- Such measures are known as non-parametric or scale-free measures of association and these are less sensitive to the distribution and scale of the data.
- Two such non-parametric measures of association are:
  a) Kendall's tau (τ)
  b) Spearman's rho (ρ_s)

Non-parametric Measures of Association

Kendall rank correlation coefficient or Kendall's tau (τ)

- Let $(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$ be the paired sample of two random variables, X and Y.
- The ordered pair (x_k, y_k) can be transformed into their respective ranks (R_k^x, R_k^y).
- Two pairs (x_i, y_i) and (x_j, y_j) are known to be
  concordant if
  $(x_i - x_j)(y_i - y_j) > 0$ or $\left(R_i^x - R_j^x\right)\left(R_i^y - R_j^y\right) > 0$
  discordant if
  $(x_i - x_j)(y_i - y_j) < 0$ or $\left(R_i^x - R_j^x\right)\left(R_i^y - R_j^y\right) < 0$.

Kendall rank correlation coefficient or Kendall's tau (τ)

- Sample estimate of Kendall's tau is obtained as the difference between the probability of concordance and the probability of discordance.
- Out of n paired samples, there are n_{C_2} different ways to select two pairs.
- If there are c number of concordant pairs and d number of discordant pairs, sample estimate of Kendall's tau is expressed as

$$\hat{\tau} = P\left[(x_i - x_j)(y_i - y_j) > 0\right] - P\left[(x_i - x_j)(y_i - y_j) < 0\right]$$

$$\hat{\tau} = \frac{c - d}{n_{C_2}}$$

Kendall rank correlation coefficient or Kendall's tau (τ)

- Kendall's tau is expected to follow normal distribution with mean 0 and variance $\frac{2(2n+5)}{9n(n-1)}$ under the assumption that X and Y are independent.
- This information can be utilised to check the significance of Kendall's tau.
- Kendall's Tau (τ) for u, v is related to copula function C, as given in the following expression:

$$\tau = 4 \int C(u,v)\, dC(u,v) - 1$$

- The above equation link the scale-free measure of association with copula functions, and hence, can be used to derive relationship between copula parameter and the scale-free measures of association.

Non-parametric Measures of Association

Spearman's rank correlation coefficient or Spearman's rho (ρ_s)

- Spearman's rank correlation coefficient (ρ_s) is analogous to correlation coefficient and calculated in similar way with one difference that instead of using the values of variables, their ranks are used.
- For a sample (x_i, y_i) of size n, to compute ρ_s the samples are first transformed to their respected ranks (R_i^x, R_i^y).
- Spearman's rho (ρ_s) is expressed as

$$\rho_s = \frac{\sum_{i=1}^{n}(R_i^x - \bar{R}^x)(R_i^y - \bar{R}^y)}{\sqrt{\sum_{i=1}^{n}(R_i^x - \bar{R}^x)^2 \sum_{i=1}^{n}(R_i^y - \bar{R}^y)^2}}$$

Spearman's rank correlation coefficient or Spearman's rho (ρ_s)

- If there is tie between two or more observations, an average of ties ranks are assigned to all those ties.
- If all the ranks are distinct, ρ_s can be computed as

$$\rho_s = 1 - \frac{6\sum_{i=1}^{n} d_i}{n(n^2-1)}$$

where $d_i = (R_i^x - R_i^y)$ and n is the number of data. R_i^x and R_i^y stand for rank of x_i and y_i in X and Y respectively.

Spearman's rank correlation coefficient or Spearman's rho (ρ_s)

- For large samples, under the assumption that X and Y are independent, Spearman's rho follows normal distribution with mean 0 and variance $\frac{1}{(n-1)}$.
- Spearman's rho between u and v is related to the copula function, as given in the following expression:

$$\rho_s = 12 \iint uv\, dC(u,v) - 3$$

- The above equation link the scale-free measure of association with copula functions, and hence, can be used to derive relationship between copula parameter and the scale-free measures of association.

Example

Let us consider X and Y as two random variables, and six paired values for these random variables are as follows:

$$(21,13), (12,17), (18,11), (15,14), (16,20), (19,27)$$

Calculate:

a) Kendall's Rank Correlation Coefficient (τ)

b) Spearman's Rank Correlation Coefficient (ρ_s).

Note: For drawing meaningful statistical inference, the data length should be sufficiently large, which is not the case in this example. This example only illustrates the procedure for calculating the scale-free measures of association. In the real world, the data set can never be this small, however the methodology for calculating the measures of association do not change and can easily be programmed using concepts from this example.

NPTEL

Solution

The given data pairs are in the increasing order of X:

X	12	15	16	18	19	21
Y	17	14	20	11	27	13

a) Calculation of Kendall's Tau (τ)

Concordant and discordant pairs can be counted for each pair and are tabulated (in the next slide).

NPTEL

Solution

Observation	Data Pairs (in the increasing order of X)	Concordant (c)	Discordant (d)
1	(12,17)	2	3
2	(15,14)	2	2
3	(16,20)	1	2
4	(18,11)	2	0
5	(19,27)	0	1
6	(21,13)	0	0
Total		7	8

Solution

Kendall's Tau $(\hat{\tau})$ is given by,

$$\hat{\tau} = \frac{c-d}{n_{C_2}} = \frac{2(c-d)}{n(n-1)} = \frac{2(7-8)}{6(6-1)}$$

$$\hat{\tau} = -0.067$$

NPTEL

Solution

b) Calculation of Spearman's Rho(ρ_s)

The ordered pair (x_k, y_k) can be transformed into their respective ranks (R_k^x, R_k^y) as shown in the table.

Original data pairs in the increasing order of X

X	12	15	16	18	19	21
Y	17	14	20	11	27	13

Ranked data

R^x	1	2	3	4	5	6
R^y	4	3	5	1	6	2

$$\bar{R}^x = \bar{R}^y = \frac{1+2+3+4+5+6}{6} = 3.5$$

Then $\sum_{i=1}^{n}(R_i^x - \bar{R}^x)^2$ and $\sum_{i=1}^{n}(R_i^y - \bar{R}^y)^2$ are computed and tabulated (in the next slide).

Solution

R_i^x	R_i^y	$(R_i^x - \bar{R}^x)$	$(R_i^x - \bar{R}^x)^2$	$(R_i^y - \bar{R}^y)$	$(R_i^y - \bar{R}^y)^2$	$(R_i^x - \bar{R}^x)(R_i^y - \bar{R}^y)$
1	4	-2.5	6.25	0.5	0.25	-1.25
2	3	-1.5	2.25	-0.5	0.25	0.75
3	5	-0.5	0.25	1.5	2.25	-0.75
4	1	0.5	0.25	-2.5	6.25	-1.25
5	6	1.5	2.25	2.5	6.25	3.75
6	2	2.5	6.25	-1.5	2.25	-3.75
$\sum$ =		0	17.5	0	17.5	-2.50

$$\sum_{i=1}^{n}(R_i^x - \bar{R}^x)^2 = \sum_{i=1}^{n}(R_i^y - \bar{R}^y)^2 = 17.5; \quad \sum_{i=1}^{n}(R_i^x - \bar{R}^x)(R_i^y - \bar{R}^y) = -2.5$$

Solution

Spearman's Rho (ρ_s) is given by,

$$\rho_s = \frac{\sum_{i=1}^{n}(R_i^x - \bar{R}^x)(R_i^y - \bar{R}^y)}{\sqrt{\sum_{i=1}^{n}(R_i^x - \bar{R}^x)^2 \sum_{i=1}^{n}(R_i^y - \bar{R}^y)^2}}$$

$$\rho_s = \frac{-2.5}{\sqrt{17.5 \times 17.5}}$$

$$\rho_s = -0.143$$





SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 05 : Introduction to Copulas and Applications

W5L4 : Commonly used Copula Functions and Selection of Best-fit Copula

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

CONCEPTS COVERED

- **Copula Functions**
- **Selection of Best-fit Copula**

NPTEL

Most Commonly Used Copula Functions

- Many copula functions exist that follow the properties of copula (discussed in previous lectures) and lie within the Frechet-Hoeffding bounds.
- Copula function can also be derived for different joint distributions on case-by case basis.
- However, several families of copula functions are commonly used in practice.
- The popular class of copula families are:
  - Elliptical copula
  - Archimedean copula

Elliptical Copula

- Elliptical copulas constitute a family of copulas derived from elliptical distributions, such as Normal distribution, Student's-t distribution etc.
- An elliptical copula tries to conserve the linear correlation between the data and it uses correlation coefficient ρ as parameter.
- The elliptical copulas do not have close formed expressions and are restricted to have a radial symmetry.
- One of the popular elliptical copulas is Gaussian copula.

Elliptical Copula

- A multidimensional (n-dimensional) Gaussian copula $C_R\ (u_1, u_2, \ldots., u_n)$ with correlation matrix R is given by

$$C_R\ (u_1, u_2, \ldots., u_n) = \Phi_R^n(\Phi^{-1}(u_1), \Phi^{-1}(u_2), \ldots. \Phi^{-1}(u_n))$$

where,

$u_1, u_2, \ldots., u_n$ represent the reduced variates,

Φ^{-1} is the inverse cumulative distribution function of a univariate standard normal distribution,

Φ_R^n is the joint cumulative distribution function of a multivariate normal distribution with zero mean vector and covariance matrix equal to correlation matrix between $\Phi^{-1}(u_1), \Phi^{-1}(u_2), \ldots. \Phi^{-1}(u_n)$, denoted as R.

Elliptical Copula

- There is no analytical, closed-form solution for this copula function
- The bivariate Gaussian copula can be written as

$$C_R\ (u, v\ ) = \Phi_R^2(\Phi^{-1}(u), \Phi^{-1}(v))$$

- In bivariate case, the correlation matrix R stands for $R\begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}$, where ρ is correlation coefficient between $\Phi^{-1}(u)$ and $\Phi^{-1}(v)$.
- The copula density for the same can be written as

$$c_R\ (u, v\ ) = \frac{1}{\sqrt{1-\rho^2}}\ exp\left\{\frac{2\rho\Phi^{-1}(u)\Phi^{-1}(v) - \rho^2(\Phi^{-1}(u)^2 + \Phi^{-1}(v)^2)}{2(1-\rho^2)}\right\}$$

Elliptical Copula

The bivariate gaussian copula can be re-written as:

$$C_R(u,v) = \int_{-\infty}^{\Phi^{-1}(u)} \int_{-\infty}^{\Phi^{-1}(v)} \frac{1}{\sqrt{1-\rho^2}} exp\left\{\frac{2\rho st - \rho^2(s^2+t^2)}{2(1-\rho^2)}\right\} dsdt$$

Another commonly used elliptical copula is *t*-copula. A 2-dimensional *t*-copula is expressed as:

$$C_{v,R}(u,v) = \int_{-\infty}^{t_v^{-1}(u)} \int_{-\infty}^{t_v^{-1}(v)} \frac{1}{2\pi\sqrt{\det(R)}} \left\{1 + \frac{s^2 - 2\rho st + t^2)}{v\det(R)}\right\}^{-(v+2)/v} dsdt$$

where R is the correlation matrix between $t_v^{-1}(u)$ and $t_v^{-1}(v)$ and 'det' represents determinant of the matrix.

Archimedean Copula

- Archimedean copula is extensively used in many fields of studies.
- By definition, any copula that can be expressed in terms of

$$C(u,v) = \phi^{[-1]}\big(\phi(u) + \phi(v)\big)$$

is known as 'Archimedean Copula', where ϕ is known as generator function of copula.

- ϕ is a convex, strictly decreasing, continuous function from [0,1] to [0,∞), such that $\phi(1) = 0$
- $\phi^{[-1]}$ is its 'pseudo-inverse' $\phi^{[-1]}$: [0,∞) → [0,1].

Archimedean Copula

- If $\phi(0)$ is ∞, then ϕ is called strict generator function and corresponding Archimedean Copula is called strict Archimedean Copula.
- In case of strict generator function, the 'pseudo inverse' ($\phi^{[-1]}$) is ϕ^{-1} , otherwise 'pseudo-inverse' ($\phi^{[-1]}$) is defined as

$$\phi^{[-1]}(t) = \begin{cases} \phi^{-1}(t), & 0 \le t \le \phi(0) \\ 0, & \phi(0) < t \le \infty \end{cases}$$

Archimedean Copula

- Any Archimedean copula C is symmetric and associative, i.e., if $u, v, w \in [0,1]$, then, $C(u,v) = C(v,u)$ and $C(C(u,v),w) = C(u,C(v,w))$.
- Furthermore, Kendall's τ and the generator function are related in case of Archimedian copula. The relationship is expressed as

$$\tau = 1 + 4\int_0^1 \frac{\phi(u)}{\phi'(u)}du$$

- This relationship is useful for obtaining joint distribution from sample measure of dependence in terms of the estimate of Kendall's tau.

Archimedean Copula

A list of commonly used Archimedean copulas is provided in following table.

Copula type	Copula function $C_\theta(u,v)$	Generator function $\phi_\theta(t)$	$\theta \in$	Kendall's tau (τ) in terms of θ
Independent	uv	$-ln(t)$		
Clayton	$[\max(u^{-\theta}+v^{-\theta}-1,0)]^{-\frac{1}{\theta}}$	$\frac{1}{\theta}(t^{-\theta}-1)$	[-1, ∞)\{0}	$\frac{\theta}{\theta+2}$
Frank	$-\frac{1}{\theta}ln\left(1+\frac{(e^{-\theta u}-1)(e^{-\theta v}-1)}{e^{-\theta}-1}\right)$	$-ln\left(\frac{e^{-\theta t}-1}{e^{-\theta}-1}\right)$	$(-\infty,\infty)\backslash\{0\}$	$1-\frac{4}{\theta}[D_1(\theta)-1]$
Ali-Mikhail-Haq	$\frac{uv}{1-\theta(1-u)(1-v)}$	$ln\left(\frac{1-\theta(1-t)}{t}\right)$	[1,-1]	$\left(\frac{3\theta-2}{\theta}\right)-\frac{2\ln(1-\theta)}{3}\left(1-\frac{1}{\theta}\right)^2$

Archimedean Copula

Copula type	Copula function $C_\theta(u,v)$	Generator function $\phi_\theta(t)$	$\theta \in$	Kendall's tau (τ) in terms of θ
Gumbel-Hougaard	$exp\left(-[(-\ln u)^\theta+(-\ln v)^\theta]^{\frac{1}{\theta}}\right)$	$(-\ln t)^\theta$	[1, ∞)	$\frac{\theta-1}{\theta}$
Joe	$1-[(1-u)^\theta+(1-v)^\theta-(1-u)^\theta(1-v)^\theta]^{\frac{1}{\theta}}$	$-log(1-(1-t)^\theta$	[1, ∞)	-

Advantages of Archimedean Copulas:

i. This class of copula can be easily constructed using generator function.

ii. There are many different Archimedean copulas available, that are applicable for a range of dependence characteristics.

iii. The different Archimedean copulas have useful properties, such as having an explicit expression based on generator, catering a high dimension using single parameter derived from the measure of dependence, etc.

Selection of Best-fit Copula

- Given the sample data, if there are more than one potential copulas, best-fit copula has to be selected.
- There are several goodness-of-fit (GOF) tests for statistically checking the suitability of a copula. Most of these approaches use:
  (a) Empirical copula
  (b) Kendall's transform.

Test using Empirical Copula

These tests compare the distance between the empirical copula $(C_n(u,v))$and parametric estimate $(C_n^{\theta}(u,v))$ of C, where u and v are the reduced variates of the sample data $X\ (x_1, x_2, \dots x_n)$ and $Y\ (y_1, y_2, \dots y_n)$ respectively, and n is the number of the observations.

NPTEL

Selection of Best-fit Copula

Test Using Empirical Copula

- The empirical copula (C_n) is defined as

$$C_n(u,v) = \frac{1}{n}\sum_{\forall u,v} \Im(U \le u, V \le v), \qquad u, v \in \mathrm{I}$$

where, $\Im(\cdot)$ is the indicator function that takes a value of 1 if the argument $(\cdot)$ is true and 0 if it is false.

- The Cramér-von Mises and Kolmogorov-Smirnov (KS) statistics are based on the distance between fitted copula and empirical copula.

NPTEL

Selection of Best-fit Copula

Test Using Empirical Copula

- The Cramér-von Mises statistic (S_n) is a popular goodness-of-fit test statistic for copula models. The statistic, S_n is expressed as

$$S_n = \sum_{\forall u,v} \left(C_n(u,v) - C_n^{\theta}(u,v)\right)^2$$

- The KS statistic (T_n) is based on the absolute maximum distance between C_n and C_n^{θ}. It is expressed as

$$T_n = \max_{u,v \epsilon I} \left|\sqrt{n}\left(C_n(u,v) - C_n^{\theta}(u,v)\right)\right|$$

- Lower the values better the fit.

NPTEL

Selection of Best-fit Copula

Test Using Kendall's Transform

- For the best-fit copula selection procedure using Kendall's transform κ is obtained from the joint distribution, derived parametrically using a particular copula, C_n^{θ}. It is expressed as

$$\kappa(t) = P\left(C_n^{\theta}(u,v) \leq t\right)$$

- The κ is determined either parametrically $\left(\kappa_n^{\theta}\right)$ or non-parametrically (κ_n). κ_n is derived using the empirical distribution function C_n as given below

$$\kappa_n(t) = \frac{1}{n} \sum_{\forall u,v} \Im\left(C_n(u,v) \leq t\right)$$

NPTEL

Selection of Best-fit Copula

Test Using Kendall's Transform

- The test statistics $\left(S_n^{(\kappa)} \; and \; T_n^{(\kappa)}\right)$ are basically the rank-based analogues of the Cramér-von Mises and KS statistics.
- The test statistic $S_n^{(\kappa)}$ and $T_n^{(\kappa)}$ are expressed as

$$S_n^{(\kappa)} = \sum_{\forall t} (\kappa_n(t) - \kappa(t))^2$$

$$T_n^{(\kappa)} = \sup_{\forall t} |\kappa_n(t) - \kappa(t)|$$

- Lower the values better the fit.

Example

Five paired observations of two random variables X and Y are given in the following table.

X	18.1	22.3	18.7	17.5	24.5
Y	22.3	26.0	25.5	30.0	28.2

The marginal distribution of X follows a normal distribution with mean of 17.5 and standard deviation of 2.7. Similarly, the marginal distribution of Y follows a normal distribution with mean 22.0 and standard deviation 4.2.

Fit the Clayton and Gumbel-Hougaard copulas to the data. Compute the Cramér-von Mises and Kolmogorov-Smirnov (KS) goodness-of-fit statistics and identify the best copula.

Solution

Given,

$X \sim N(17.5, 2.7)$

$Y \sim N(22.0, 4.2)$

Conversion of the observations into reduced variates:

X and Y can be converted to reduced variates (u, v) through their respective marginal distributions (standard normal table is provided in the next slide).

$$u = F_X(x) = \Phi\left(\frac{x - \mu_X}{\sigma_X}\right)$$

$$v = F_Y(y) = \Phi\left(\frac{y - \mu_Y}{\sigma_Y}\right)$$

Solution

Thus the reduced variates u and v are obtained as follows:

X	Y	u	v
18.1	22.3	0.588	0.622
22.3	26.0	0.962	0.830
18.7	25.5	0.672	0.798
17.5	30.0	0.500	0.972
24.5	28.2	0.995	0.930

Tables of the Normal Distribution

Probability Content from -oo to Z

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

Solution

Kendall's rank correlation

Two pairs (x_i, y_i) and (x_j, y_j) are known to be concordant if $(x_i - x_j)(y_i - y_j) > 0$ or $\left(R_i^x - R_j^x\right)\left(R_i^y - R_j^y\right) > 0$ and discordant if $(x_i - x_j)(y_i - y_j) < 0$ or $\left(R_i^x - R_j^x\right)\left(R_i^y - R_j^y\right) < 0$.

X	Y	R_X	R_Y	R_X, R_y	Concordant (c)	Discordant (d)
17.5	30.0	1	5	(1,5)	0	4
18.1	22.3	2	1	(2,1)	3	0
18.7	25.5	3	2	(3,2)	2	0
22.3	26.0	4	3	(4,3)	1	0
24.5	28.2	5	4	(5,4)	0	0
Total					6	4

Solution

Kendall's rank correlation

Number of concordant pairs = 6, Number of discordant pairs = 4

Hence,

$$\tau = \frac{c-d}{n_{C_2}} = \frac{2(c-d)}{n(n-1)} = \frac{2(6-4)}{5(5-1)} = 0.2$$

Solution

Clayton Copula Fitting

The parameter for Clayton copula can be calculated from

$$\tau = \frac{\theta}{\theta + 2}$$

$$\text{or, } 0.2 = \frac{\theta}{\theta + 2}$$

$$\text{or, } 0.2(\theta + 2) = \theta$$

$$\theta = 0.5$$

The value of θ is valid for Clayton copula because the parameter for Clayton copula must be in the range of $[-1, \infty) \setminus \{0\}$.

NPTEL

Solution

Clayton Copula Fitting

The Clayton copula is $C_\theta^n(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-\frac{1}{\theta}}$

Observation	$u^{-0.5}$	$v^{-0.5}$	$u^{-0.5} + v^{-0.5} - 1$	$C_\theta^n(u, v)$
1	1.304	1.268	1.573	0.404
2	1.019	1.098	1.117	0.801
3	1.220	1.120	1.340	0.557
4	1.414	1.014	1.429	0.490
5	1.002	1.037	1.039	0.926

Solution

Gumble-Hougaard Copula Fitting:

The parameter for Gumble-Hougaard copula can be calculated from

$$\tau = \frac{\theta - 1}{\theta}$$

$$\text{or, } 0.2 = \frac{\theta - 1}{\theta}$$

$$\text{or, } 0.2\theta = \theta - 1,$$

$$\theta = 1.25$$

The value of θ =1.25 is valid Gumble-Hougaard copula ($\theta \in [1, \infty)$.

NPTEL

Solution

Gumble-Hougaard Copula Fitting:

The Gumble-Hougaard copula is expressed as

$$C_\theta^n(u, v) = exp\left(-[((-\ln u)^\theta + (-\ln v)^\theta)]^{\frac{1}{\theta}}\right)$$

Observation	$(-\ln u)$	$(-\ln v)$	$\left[\left((-\ln u)^\theta + (-\ln v)^\theta\right)\right]^{\frac{1}{\theta}}$	$C_\theta^n(u, v)$
1	0.453	0.395	0.704	0.416
2	0.017	0.123	0.207	0.813
3	0.316	0.156	0.548	0.578
4	0.632	0.012	0.644	0.495
5	0.001	0.038	0.075	0.928

Solution

Calculation of Empirical Copula

For a sample of size n , the empirical copula at the point (u, v) is

$$C_n(u,v) = \frac{1}{n}\sum_{\forall u,v} \Im(U \le u, V \le v),$$

Observation	u	v	$C_n(u,v)$
1	0.588	0.622	0.2
2	0.962	0.830	0.6
3	0.672	0.798	0.4
4	0.500	0.972	0.2
5	0.995	0.930	0.8

Solution

Best fit copula

The Cramér-von Mises statistic is given by $S_n = \sum_{\forall u,v} \left(C_n(u,v) - C_n^{\theta}(u,v)\right)^2$

Observation	$C_n(u,v)$	$C^n_{\theta,Clayton}$	$C^n_{\theta,Gumble}$	$\left(C_n(u,v) - C^n_{\theta,Clayton}\right)^2$	$\left(C_n(u,v) - C^n_{\theta,Gumble}\right)^2$
1	0.2	0.404	0.416	0.0416	0.0466
2	0.6	0.801	0.813	0.0404	0.0453
3	0.4	0.557	0.578	0.0246	0.0316
4	0.2	0.490	0.495	0.0841	0.08702
5	0.8	0.926	0.928	0.0158	0.0163
Total				0.207	0.227

Solution

Best fit copula

The KS statistic is given by $T_n = \max_{u,v \epsilon I} \left| \sqrt{n} \left(C_n(u,v) - C_n^{\theta}(u,v) \right) \right|$

Observation	$C_n(u,v)$	$C^n_{\theta,Clayton}$	$C^n_{\theta,Gumble}$	$(C_n(u,v) - C^n_{\theta,Clayton})$	$(C_n(u,v) - C^n_{\theta,Gumble})$
1	0.2	0.404	0.416	-0.204	-0.216
2	0.6	0.801	0.813	-0.201	-0.213
3	0.4	0.557	0.578	-0.157	-0.178
4	0.2	0.490	0.495	-0.290	-0.295
5	0.8	0.926	0.928	-0.126	-0.128
$max\left\|\sqrt{n}(C_n(u,v) - C^n_{\theta})\right\|$				0.648	0.659

Solution

However, in some applications, S_n and T_n may identify different best-fitting copulas. In such cases, the copula selected based on S_n is generally preferred because S_n has higher statistical power than T_n.

In this problem, lower values of both S_n (for Clayton 0.207 and Gumble 0.227) and T_n (for Clayton 0.648 and Gumble 0.659) suggest that Clayton copula is the best fit copula compared to Gumbel-Hougaard copula.


FREE ONLINE EDUCATION
swayam
Thank You!



SWAYAM NPTEL

ONLINE CERTIFICATION COURSE ON

Statistical and Machine Learning Methods: Basics and Applications

MODULE 05 : Introduction to Copulas and Applications

W5L5: Applications of Copulas

BY

Prof. Rajib Maity

Professor, Department of Civil Engineering
Chairperson, School of Water Resources
Indian Institute of Technology Kharagpur
Kharagpur – 721302, West Bengal

1

CONCEPTS COVERED

- Use of Copulas
- Applications of Copulas in Machine Learning

NPTEL

2

Use of Copulas

- Copula can be used in many studies for
  - data generation,
  - multivariate frequency analysis,
  - probabilistic prediction
  - machine learning approaches

  and many other applications.

NPTEL

Data Generation

- Data generation using copula preserves the dependence structure between the associated variables.
- Two methods exist for data generation using copulas –
  - one is specific to Archimedean copula, and
  - other can be applied to an copula.
- These methods are discussed one after another.

NPTEL

Data Generation: using Archimedean Copulas

- For an Archimedean copula, functional forms of $\phi^{[-1]}(\bullet), \phi'(\bullet)$ and $\phi'^{[-1]}(\bullet)$ are obtained using $\phi_\theta(\bullet)$, which is the generator function with parameter θ. The following equation is used to obtain $\phi^{[-1]}(\bullet)$. Same can be used for $\phi'(\bullet)$, which is derivative of $\phi(\bullet)$ with respect to •.

$$\phi^{[-1]}(t) = \begin{cases} \phi^{-1}(t), & 0 \le t \le \phi(0) \\ 0, & \phi(0) < t \le \infty \end{cases}$$

- Two independent uniformly distributed $(\mathrm{U}(0,1))$ random variates, u and r, are generated.
- Two new variables, S and W, are obtained as

$$s = \phi'(u)/r \text{ and } w = \phi'^{[-1]}(s)$$

Data Generation: using Archimedean Copulas

- Another variable, v, is obtained as

$$v = \phi^{-1}(\phi(w) - \phi(u))$$

- The pair u and v are the simulated pair, preserving the dependence structure.
- Both these u and v are in the range $[0, 1]$.
- These simulated pairs of u and v are then back transformed through their corresponding cumulative marginal distributions.

Data Generation: General Procedure using any Copula

- The value of u is substituted in the expression for conditional distribution of v given u.
- The expression is equated to p to solve for v.
- This gives the values of v based on the dependence structure between u and v.
- Both u and v are in the range $[0, 1]$.
- These simulated pairs of u and v are then back transformed using their corresponding inverse cumulative marginal distribution.

Example

From the historical records, monthly maximum rainfall duration (in hour) is found to follow exponential distribution with mean maximum rainfall duration of half an hour. The monthly maximum discharge (in cumec) is found to follow a normal distribution with mean 500 cumec and standard deviation of 36.5 cumec. The joint distribution between these variables is obtained by using Ali-Mikhail-Haq copula with $\theta = 0.5$.

Generate monthly maximum rainfall duration and monthly maximum discharge for a year.

Solution

Let us assume that X and Y are two random variables (with corresponding reduced variates u and v) representing the monthly maximum rainfall duration and monthly maximum discharge respectively. The CDF for X and Y is given by

$$F_X(x) = 1 - e^{-\lambda x} = 1 - e^{-2x} \qquad x \geq 0$$

$$F_Y(y) = \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy \qquad -\infty < y < \infty$$

$$= \int_{-\infty}^{y} \frac{1}{91.49} e^{-\frac{(y-500)^2}{2664.5}} dy$$

Solution

The joint distribution of variables X and Y, using Ali-Mikhail-Haq copula with $\theta = 0.5$ can be evaluated using

$$F_{X,Y}(x,y) = \frac{uv}{1-\theta(1-u)(1-v)} = \frac{uv}{1-0.5(1-u)(1-v)}$$

The conditional distribution for Y conditioned on X is given by

$$F_{Y/X}(y/X = X) = \frac{\partial F_{X,Y}(x,y)}{\partial u}$$

$$= \frac{uv\left(\frac{v}{2}-\frac{1}{2}\right)}{\left(\left(\frac{u}{2}-\frac{1}{2}\right)(v-1)-1\right)^2} - \frac{v}{\left(\frac{u}{2}-\frac{1}{2}\right)(v-1)-1}$$

Solution

The 12 randomly generated values for

$$u = [0.74, 0.08, 0.13, 0.10, 0.97, 0.33, 0.56, 0.03, 0.20, 0.70, 0.04, 0.94]$$

Similarly,12 random values for

$$p = [0.44, 0.38, 0.77, 0.80, 0.19, 0.49, 0.45, 0.65, 0.71, 0.75, 0.63, 0.25]$$

For estimating the values of v, the conditional distribution of Y given X is solved by substituting u and equating to p.

The obtained values of v are

$$v = [0.51, 0.26, 0.68, 0.71, 0.29, 0.45, 0.48, 0.50, 0.63, 0.79, 0.48, 0.35]$$

Solution

The reduced variates u and v can be transformed back to the variables X and Y using the relationship given above. Hence, the back transformed variables are given by

$$X = [0.67, 0.04, 0.07, 0.05, 1.75, 0.20, 0.41, 0.02, 0.11, 0.60, 0.02, 1.41]$$

$$Y = [501.0, 476.8, 517.0, 519.8, 479.7, 495.4, 497.9, 499.6, 512.4, 529.0, 498.1, 486.4]$$

Probabilistic Prediction Using Copulas

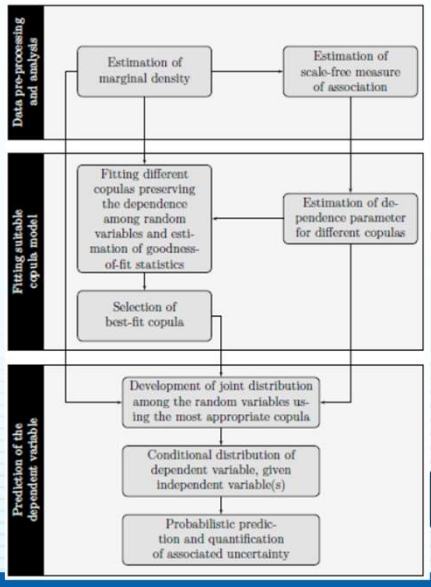
Another potential application of copulas is probabilistic prediction of random variables. Major steps to be followed are presented in a flow chart .

The three major steps are:

(A) Data pre-processing and analysis

(B) Fitting suitable copula model

(C) Prediction of the dependent variable.



13

Multivariate Frequency Analysis using Copulas

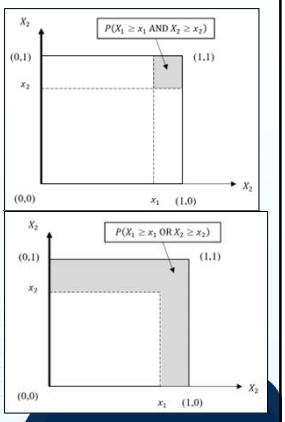
- The joint distribution between two or more variables, as developed through copulas, can be used for bivariate and multivariate frequency analysis.
- The concept of frequency analysis is extended to the case of two or more than two dimensional joint distribution.
- Such distribution is used to calculate joint probability of exceedance that leads to the concept of Joint Return Period (JRP).
- Copulas capture the dependence between variables, enabling more reliable estimation of joint probabilities and joint return periods for bivariate and multivariate case.

14

Frequency Analysis using Copula

Bivariate case

- For the bivariate case, let assume two variables X_1 and X_2, and as obtained through copula functions, their joint cumulative distribution is $F_{X_1,X_2}(x_1, x_1)$.
- Now, occurrences of a particular event, say $X_1 \geq x_1$ can be combined with another event, say $X_2 \geq x_2$ through 'AND' or 'OR' conditions
- Here, 'AND' means both the events have occurred simultaneously, and 'OR' mean at least one of them has occurred.



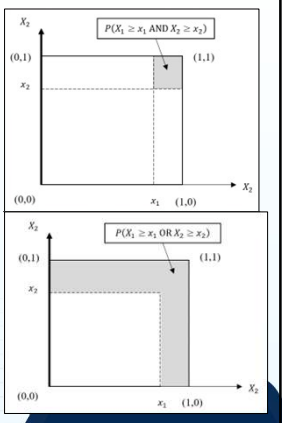
Frequency Analysis using Copula

Bivariate case

Now, JRP can be calculated for both the 'AND' or 'OR' conditions using copula, denoted as T_{AND} and T_{OR}, respectively.

$$T_{AND} = \frac{\mu}{1 - u_1 - u_2 + C(u_1, u_2)} \qquad T_{OR} = \frac{\mu}{1 - C(u_1, u_2)}$$

where u_1 denotes the marginal cumulative probability (CDF) of the first variable and u_2 denotes the marginal cumulative probability (CDF) of the second variable and μ is average recurrence interval of observations.



Frequency Analysis using Copula

Trivariate case (Three variables)

$$T_{AND} = \frac{\mu}{P(X_1 \geq x_1 \text{ AND } X_2 \geq x_2 \text{ AND } X_3 \geq x_3)} = \frac{\mu}{1 - u_1 - u_2 - u_3 + C(u_1, u_2) + C(u_1, u_3) + C(u_2, u_3) - C(u_1, u_2, u_3)}$$

$$T_{OR} = \frac{\mu}{P(X_1 \geq x_1 \text{ OR } X_2 \geq x_2)} = \frac{\mu}{1 - C(u_1, u_2, u_3)}$$

- Similarly, equations for higher dimensions can be developed to estimate the corresponding joint return periods.

Applications of Copulas in Machine Learning

- Copulas solve a fundamental bottleneck in machine learning while modeling complex, multivariate distributions without being forced into restrictive, homogeneous assumptions like multivariate normality.
- Copula-Neural Networks use a neural network as a feature extractor, but the final layer outputs the parameters of a specific copula family (e.g., Clayton, Gumbel, or a vine copula) and its marginal distributions.
- Copula Bayesian Networks (CBNs) utilize copulas to define the edges (dependencies) between nodes. This allows every node in the network to have a completely different continuous marginal distribution (e.g., Gamma, Lognormal, Weibull) while the copulas handle the multivariate joint distribution.

Applications of Copulas in Machine Learning

- Anomaly detection: Standard anomaly detection in machine learning often looks for marginal outliers, i.e., the points that are extreme in one or more individual dimensions.
- However, ML models can isolate the pure dependence structure by mapping multi-dimensional data into the uniform copula space, i.e. I^d.
- Anomalies can then be detected by identifying instances that violate the learned copula density, even if the individual marginal values appear completely normal.
- Detecting compound extreme events, such as a simultaneous agricultural drought and heatwave, where neither the lack of precipitation nor the high temperature is a historical outlier on its own, but their concurrent combination breaks standard dependency rules.

